

Notation follows of Goodfellow et al
Deep Learning, but handwritten is slightly
different

 a

scalar

 S $\vec{a}$

vector

 A

matrix

 $\mathcal{A}$

tensor

(will distinguish in
context)

Scalars

$$a = 5$$

Number. can be drawn from the real line

$a \in \mathbb{R}$ means any real number.

Corresponds to a floating point value
in Python, C, ...

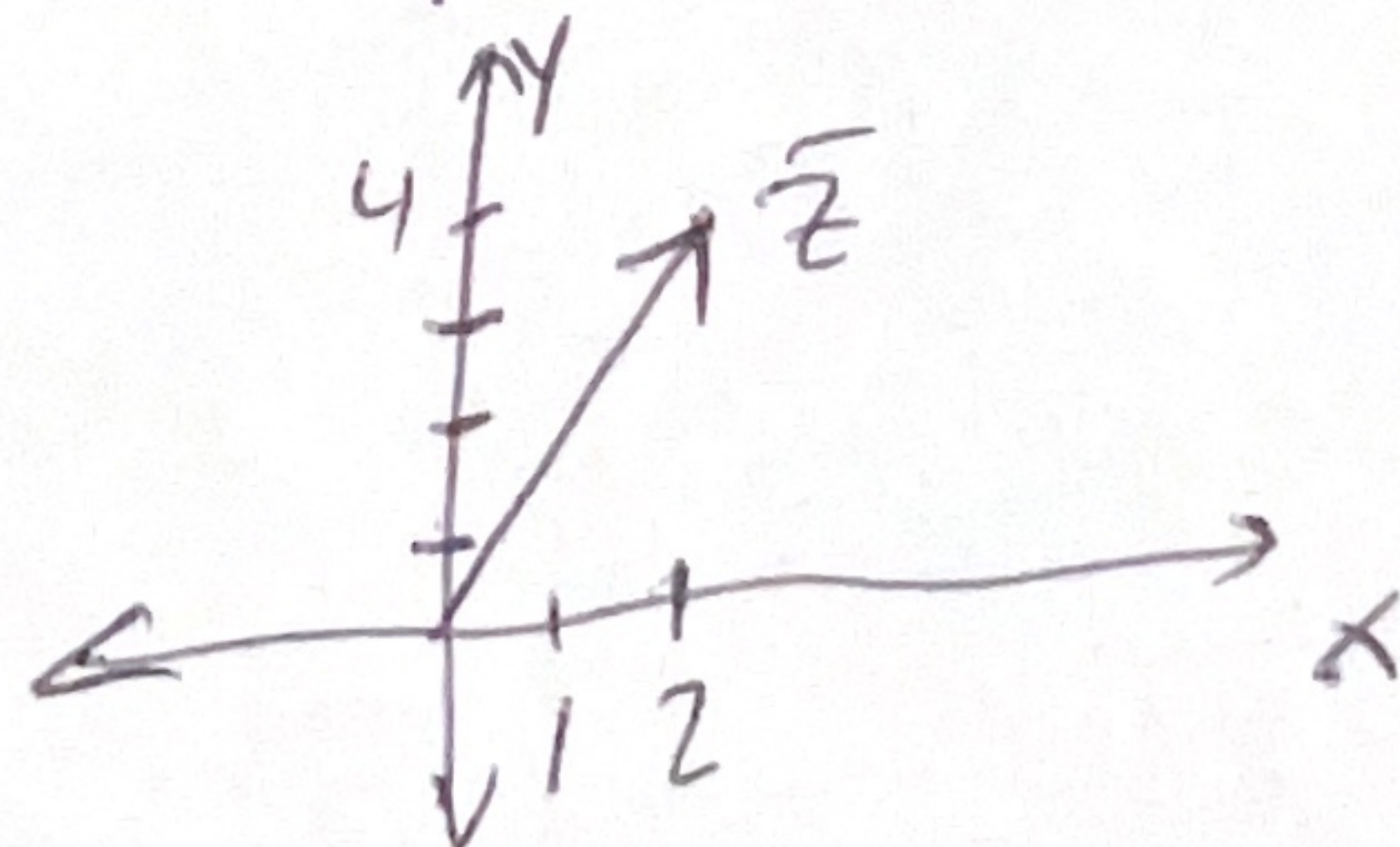
Vector

$$\bar{x} = (1, 3, 6)$$

$$\bar{x} = \begin{bmatrix} 1 \\ 3 \\ 6 \end{bmatrix}$$

← column vector, also
a simple example of a
matrix

$$\bar{z} = \begin{bmatrix} 2 \\ 4 \end{bmatrix}$$



see jupyter notebook in repository
at lecture 02/CSCI 632 - Linear Algebra - Examples.ipynb

Matrices
 → Given a 2x2 matrix A

Matrices

$$\begin{bmatrix} A_{1,1} & A_{1,2} \\ A_{2,1} & A_{2,2} \end{bmatrix} = \begin{bmatrix} 5 & 3 \\ 2 & 4 \end{bmatrix}. \quad A_{2,1} \text{ is } 2.$$

In python using numpy $A_{1,1}$ would be referenced using 0 indexing. Therefore,

$$A = \text{numpy.array}([[5, 3], [2, 4]])$$

`print(A[1,0])`

output:

2

subtract 1 from matrix ^{row} index to ~~ob~~ obtain numpy array row index.

Tensors

- A tensor is a matrix with any number of dimensions
- A ~~matrix~~ matrix is a 2-d tensor.
- A vector is a 1-d tensor.
- A scalar is a 0-d tensor.

Matrix operations

Transpose

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 5 \end{bmatrix}$$

$$A^T = \begin{bmatrix} 1 & 3 \\ 2 & 5 \end{bmatrix}$$

transpose

Transpose flips elements around the diagonal.

$$B = \begin{bmatrix} 7 & 8 & 1 \\ 5 & 6 & -1 \\ 2 & 3 & 4 \end{bmatrix}$$

$$B^T = \begin{bmatrix} 7 & 5 & 2 \\ 8 & 6 & 3 \\ 1 & -1 & 4 \end{bmatrix}$$

Adding column vectors

$$\begin{bmatrix} 1 \\ 2 \end{bmatrix} + \begin{bmatrix} 3 \\ 4 \end{bmatrix} = \begin{bmatrix} 4 \\ 6 \end{bmatrix}$$

Adding matrices

$$\begin{bmatrix} 1 & 2 \\ 3 & 5 \end{bmatrix} + \begin{bmatrix} 4 & -1 \\ 1 & -6 \end{bmatrix} = \begin{bmatrix} 5 & 1 \\ 4 & -1 \end{bmatrix}$$

$$\begin{bmatrix} 7 & 8 & 1 \\ 5 & 6 & -1 \\ 2 & 3 & 4 \end{bmatrix} + \begin{bmatrix} 1 & 2 \\ 3 & 5 \end{bmatrix} = ?$$

Meaningless! Invalid operation.

We can only add matrices that have the same dimensions, i.e., the same size in each dimension.

Subtraction

$$\begin{bmatrix} 1 & 2 \\ 3 & 5 \end{bmatrix} - \begin{bmatrix} 0 & 1 \\ 1 & 2 \end{bmatrix} = \begin{bmatrix} 1-0 & 2-1 \\ 3-1 & 5-2 \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 2 & 3 \end{bmatrix}$$

Inner Product, $\vec{a} \cdot \vec{b}$, dot product where $\vec{a} = (1, 4, 2)$, $\vec{b} = (-2, 0, 3)$

$$(1, 4, 2) \cdot (-2, 0, 3) = (1 \cdot -2) + (4 \cdot 0) + (2 \cdot 3) = -2 + 6 = 4$$

The result is a scalar.

Using matrix representation, we can rewrite the vectors as column vectors.

$$\begin{bmatrix} 1 \\ 4 \\ 2 \end{bmatrix} \cdot \begin{bmatrix} -2 \\ 0 \\ 3 \end{bmatrix} = \vec{a}^T \vec{b} = [1 \ 4 \ 2] \begin{bmatrix} -2 \\ 0 \\ 3 \end{bmatrix}$$

$$= (1 \cdot -2) + (4 \cdot 0) + (2 \cdot 3) = 4$$

Matrix Multiplication

can be viewed as a series of dot product.

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 5 \end{bmatrix}, \quad B = \begin{bmatrix} -2 & 1 \\ 4 & 0 \end{bmatrix}$$

$$\text{Let } C = AB$$

$$C_{ij} = i^{\text{th}} \text{ row of } A \cdot j^{\text{th}} \text{ column of } B$$

C_{ij} = i^{th} row and j^{th} column of matrix C .

$$C_{11} = \underbrace{[1 \quad 2]}_{i=1 \text{ row from A}} \underbrace{\begin{bmatrix} 7 \\ 4 \end{bmatrix}}_{j=1 \text{ column from B}} = 1 \cdot 7 + 2 \cdot 4 = 7 + 8 = 15$$

$$\begin{bmatrix} 1 & 2 \\ 3 & 5 \end{bmatrix} \begin{bmatrix} 7 & 1 \\ 4 & 0 \end{bmatrix} = \begin{bmatrix} 1 \cdot 7 + 2 \cdot 4 & - \\ - & - \end{bmatrix} = \begin{bmatrix} 15 & - \\ - & - \end{bmatrix}$$

$$\begin{bmatrix} 1 & 2 \\ 3 & 5 \end{bmatrix} \begin{bmatrix} 7 & 1 \\ 4 & 0 \end{bmatrix} = \begin{bmatrix} 15 & - \\ 3 \cdot 7 + 5 \cdot 4 & - \end{bmatrix} = \begin{bmatrix} 15 & - \\ 41 & - \end{bmatrix}$$

$$\begin{bmatrix} 1 & 2 \\ 3 & 5 \end{bmatrix} \begin{bmatrix} 7 & 1 \\ 4 & 0 \end{bmatrix} = \begin{bmatrix} 15 & 1 \cdot 1 + 2 \cdot 0 \\ 41 & - \end{bmatrix} = \begin{bmatrix} 15 & 1 \\ 41 & - \end{bmatrix}$$

$$\begin{bmatrix} 1 & 2 \\ 3 & 5 \end{bmatrix} \begin{bmatrix} 7 & 1 \\ 4 & 0 \end{bmatrix} = \begin{bmatrix} 15 & 1 \\ 41 & 3 \cdot 1 + 5 \cdot 0 \end{bmatrix} = \begin{bmatrix} 15 & 1 \\ 41 & 3 \end{bmatrix}$$

What is the time complexity of a naive implementation?
Given two $n \times n$ matrices A & B .

Find: Time complexity

$n \cdot n = n^2$ entries in the matrix.

Each entry is a dot product of the i^{th} row of A with the j^{th} column of B .

Dot product n multiplications followed by $n-1$ additions. $T_{\text{dot product}} = 2n-1$

$$T = n^2 \cdot (2n-1) = O(n^3) \text{ operations.}$$

Does $AB = BA$?

$$BA = ?$$

$$\begin{bmatrix} 7 & 1 \\ 0 & 4 \\ 4 & 0 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 3 & 5 \end{bmatrix} = \begin{bmatrix} 7 \cdot 1 + 1 \cdot 3 & 7 \cdot 2 + 1 \cdot 5 \\ 4 \cdot 1 + 0 \cdot 3 & 4 \cdot 2 + 0 \cdot 5 \end{bmatrix}$$

$$= \begin{bmatrix} 10 & 19 \\ 4 & 8 \end{bmatrix}$$

In this case $AB \neq BA$

What if $A = A^T$ and $B = B^T$?

Means A & B are symmetric matrices.

They are symmetric about the diagonal.

$$\begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix} \begin{bmatrix} 3 & -1 \\ -1 & 7 \end{bmatrix} = \begin{bmatrix} 1 & 13 \\ 2 & 26 \end{bmatrix}$$

$$\uparrow 2 \cdot (-1) + 4 \cdot 7$$

$$\begin{bmatrix} 3 & -1 \\ -1 & 7 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix} = \begin{bmatrix} 1 & 2 \\ 13 & 26 \end{bmatrix}$$

When A & B are symmetric $AB = (BA)^T$

~~$A^T B$~~

$$AB = (B^T A^T)^T$$

$$\begin{matrix} A & B & & B^T A^T \\ \begin{bmatrix} 1 & 2 \\ 3 & 5 \end{bmatrix} & \begin{bmatrix} 7 \\ 4 \end{bmatrix} & \begin{bmatrix} 1 & 7 \\ 0 & 6 \end{bmatrix} & \Rightarrow & \begin{bmatrix} 7 & 4 \\ 1 & 0 \end{bmatrix} & \begin{bmatrix} 1 & 3 \\ 2 & 5 \end{bmatrix} \\ & & & & & \end{matrix}$$

$$\begin{bmatrix} 1 \cdot 7 + 2 \cdot 4 \\ 15 \end{bmatrix}$$

$$\begin{bmatrix} 7 \cdot 1 + 4 \cdot 2 \\ 15 \end{bmatrix}$$

$$\begin{bmatrix} 15 \end{bmatrix}$$

If we just look at the diagonal,

diagonal of AB = diagonal of $B^T A^T$

What about off diagonal?

$$\begin{matrix} A & B & & B^T & A^T \\ \begin{bmatrix} 1 & 2 \\ 3 & 5 \end{bmatrix} & \begin{bmatrix} 7 \\ 4 \end{bmatrix} & \begin{bmatrix} 1 & 7 \\ 0 & 6 \end{bmatrix} & \Rightarrow & \begin{bmatrix} 7 & 4 \\ 1 & 0 \end{bmatrix} & \begin{bmatrix} 1 & 3 \\ 2 & 5 \end{bmatrix} \\ & & & & (B^T & A^T)^T \end{matrix}$$

However

$$\begin{matrix} A & B & & (B^T & A^T)^T \\ \begin{bmatrix} 1 & 2 \\ 3 & 5 \end{bmatrix} & \begin{bmatrix} 7 \\ 4 \end{bmatrix} & \begin{bmatrix} 1 & 7 \\ 0 & 6 \end{bmatrix} & \Rightarrow & \begin{bmatrix} 7 & 4 \\ 1 & 0 \end{bmatrix} & \begin{bmatrix} 1 & 3 \\ 2 & 5 \end{bmatrix} \end{matrix}$$

Multiplication Properties

$$A(B+C) = AB + AC$$

$$\begin{bmatrix} 1 & 2 \\ 3 & 5 \end{bmatrix} \left(\begin{bmatrix} 4 & 6 \\ 7 & 9 \end{bmatrix} + \begin{bmatrix} -1 & -2 \\ 8 & 0 \end{bmatrix} \right)$$

$$\begin{bmatrix} 1 & 2 \\ 3 & 5 \end{bmatrix} \begin{bmatrix} 4-1 & 6-2 \\ 7+8 & 9+0 \end{bmatrix} = \begin{bmatrix} 1 \cdot (4-1) + 2(7+8) & - \\ - & - \end{bmatrix}$$

$$\begin{bmatrix} 1 & 2 \\ 3 & 5 \end{bmatrix} \begin{bmatrix} 4 & 6 \\ 7 & 9 \end{bmatrix} + \begin{bmatrix} 1 & 2 \\ 3 & 5 \end{bmatrix} \begin{bmatrix} -1 & -2 \\ 8 & 0 \end{bmatrix}$$

$$\begin{bmatrix} 1 \cdot 4 + 2 \cdot 7 & - \\ - & - \end{bmatrix} + \begin{bmatrix} 1 \cdot -1 + 2 \cdot 8 & - \\ - & - \end{bmatrix}$$

$$= \begin{bmatrix} 1 \cdot (4-1) + 2(7+8) & - \\ - & - \end{bmatrix}$$

Outer Product

Less common than inner product.

Inner product produces a scalar.

Outer product is also between two vectors,

but it produces a matrix.

Let $\vec{a} = (1, 7)$, $\vec{b} = (-1, 2)$

$\swarrow x_a \searrow y_a$ $\swarrow x_b \searrow y_b$

$$\vec{a} \otimes \vec{b} = \begin{bmatrix} 1 \cdot -1 & 1 \cdot 2 \\ 7 \cdot -1 & 7 \cdot 2 \end{bmatrix} = \begin{bmatrix} x_a x_b & x_a y_b \\ y_a x_b & y_a y_b \end{bmatrix}$$

If we represent $\vec{a}$ & $\vec{b}$ as column vectors.

$$\vec{a} \otimes \vec{b} = \vec{a} \vec{b}^T$$

$$= \begin{bmatrix} 1 \\ 7 \end{bmatrix} \begin{bmatrix} -1 & 2 \end{bmatrix}$$

row 1 of $A = 1$

column 1 of $B = -1$

so

$$\begin{bmatrix} 1 \\ 7 \end{bmatrix} \begin{bmatrix} -1 & 2 \end{bmatrix} = \begin{bmatrix} 1 \cdot -1 & - \\ 7 \cdot -1 & - \end{bmatrix}$$

Hadamard Product

a.k.a., element-wise product

$$A \odot B$$

↑ Hadamard product operator.

$$\text{Let } C = A \odot B$$

$$C_{ij} = A_{ij} \cdot B_{ij}$$

Ex:

$$A = \begin{bmatrix} 5 & 13 & 7 \\ 19 & 2 & 11 \end{bmatrix}, \quad B = \begin{bmatrix} 4 & 9 & 21 \\ 6 & 14 & 3 \end{bmatrix}$$

$$A \odot B = \begin{bmatrix} 5 \cdot 4 & 13 \cdot 9 & 7 \cdot 21 \\ 19 \cdot 6 & 2 \cdot 14 & 11 \cdot 3 \end{bmatrix} = \begin{bmatrix} 20 & 117 & 147 \\ 114 & 28 & 33 \end{bmatrix}$$

Does come up in backpropagation.
We'll come back to this.

Ways of using matrices...

- commonly used to represent a system of equations

$$A \bar{x} = \bar{b}$$

$$\begin{bmatrix} 1 & 0 & 3 \\ 0 & 4 & 2 \\ 5 & 1 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 9 \\ 10 \\ 6 \end{bmatrix}$$

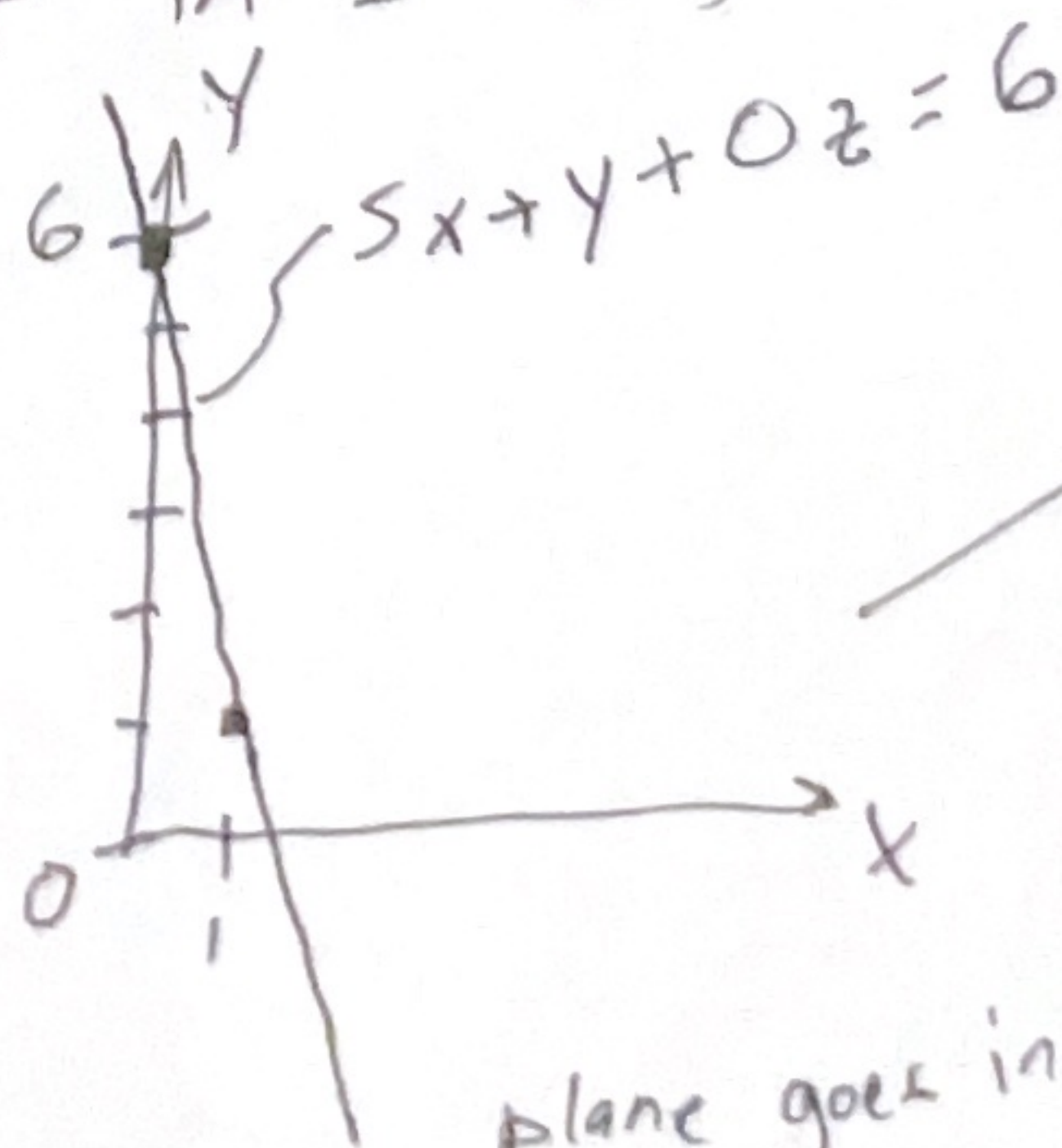
same as

$$1x + 0y + 3z = 9$$

$$0x + 4y + 2z = 10$$

$$5x + 1y + 0z = 6$$

Each row of A can be viewed as a hyperplane where the corresponding element in $\bar{b}$ offsets the hyperplane from the origin. i^{th} row is the i^{th} hyperplane and i^{th} element in $\bar{b}$ defines its offset from the origin.



plane goes in and out of the plane of the paper.